

# MONSOON/FORESTS



## ● Factors affecting climate of India

1. Latitude
2. Altitude
3. Pressure and wind system
4. Relief features
5. Ocean Currents
6. Distance from Sea

● Climate: the average weather in a given area over a longer period of time. Data taken of 30 years

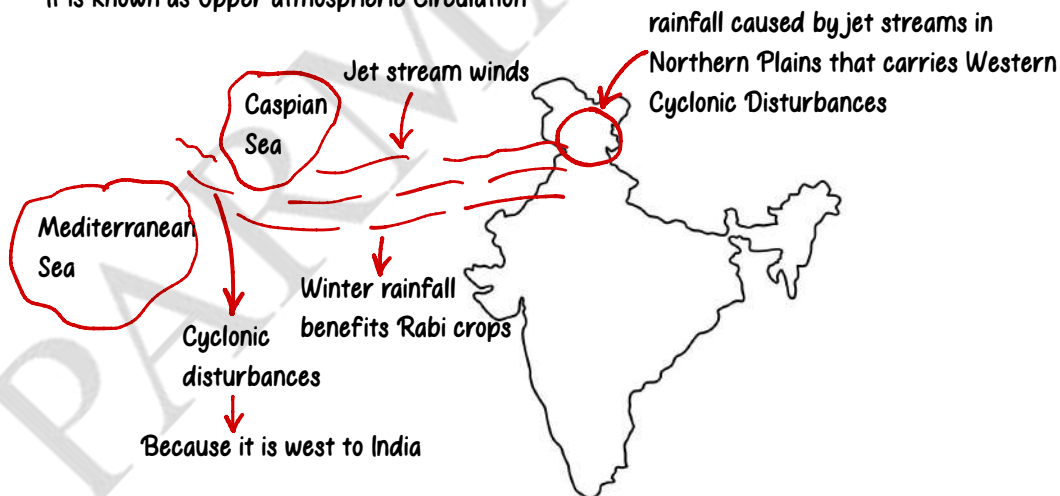
● Weather: the term refers to temporary conditions of the atmosphere, including temperature, atmospheric pressure, wind, humidity, precipitation, and cloud cover

### Winter Season

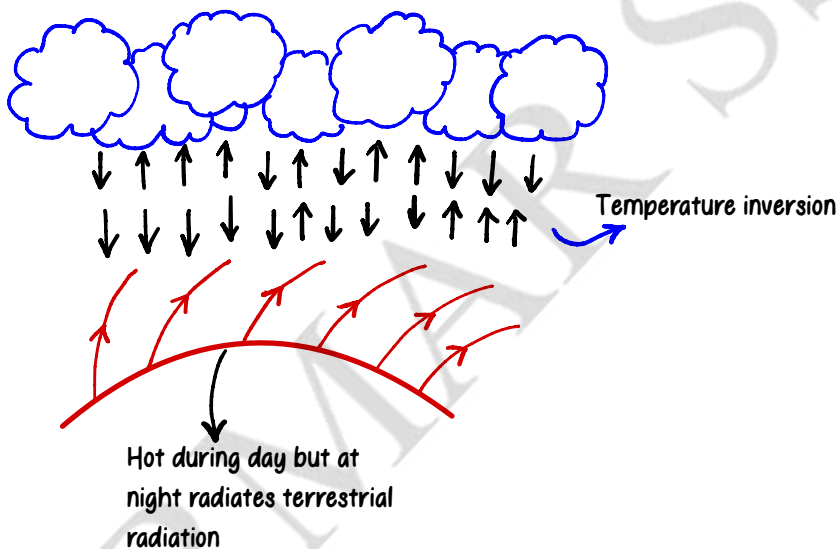
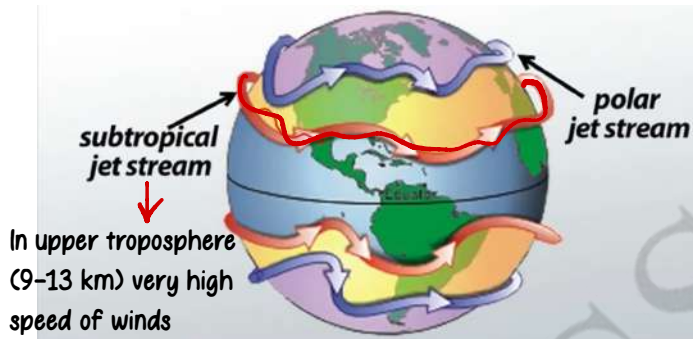
Chilling winds from Arctic/Central Asia affects winter season

### Winter Rainfall

It is known as Upper atmospheric Circulation



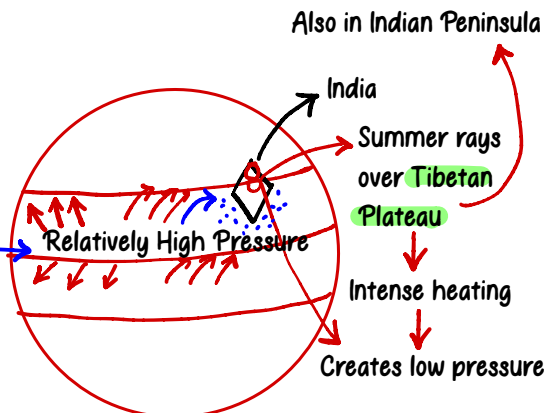
Inward of Western Disturbances through jet streams is marked by increase in prevailing night temperature



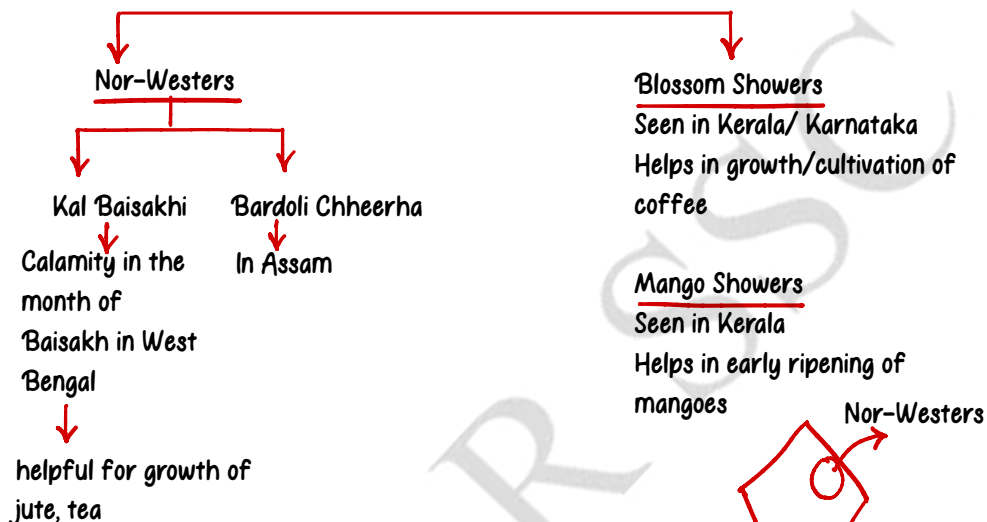
### Summer Season

ITCZ now shifts in upper region

- Trade winds meet and air ascends
- During July: 20°-25°N



## Pre-Monsoon Showers



### • On Set of Monsoon

It is derived from Arabic word "Mausim" meaning seasonal reversal of winds

Beginning in Kerala in 1st week of June in the Western Ghats

Orographic rainfall

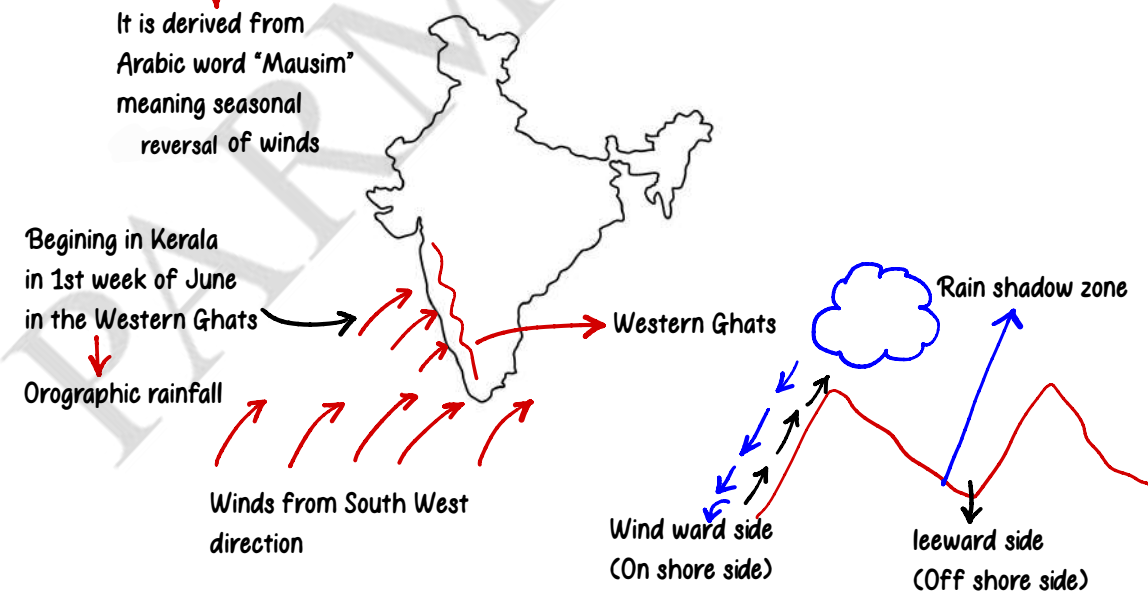
Winds from South West direction

Western Ghats

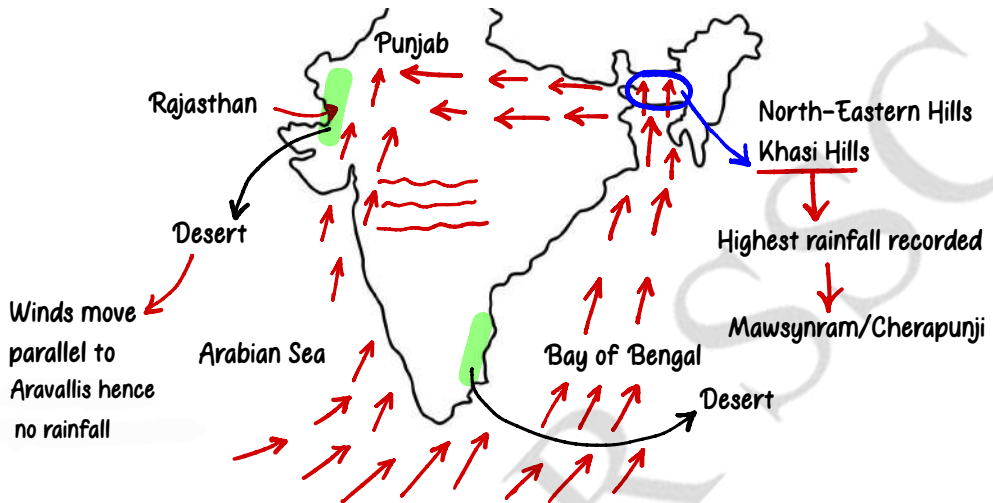
Wind ward side  
(On shore side)

Rain shadow zone

leeward side  
(Off shore side)



## Rain Bearing System



- Bay of Bengal branch and Arabian Sea branch of South West Monsoon meet at Punjab

### Break in Monsoon

Sudden decrease in rainfall after the onset of monsoon

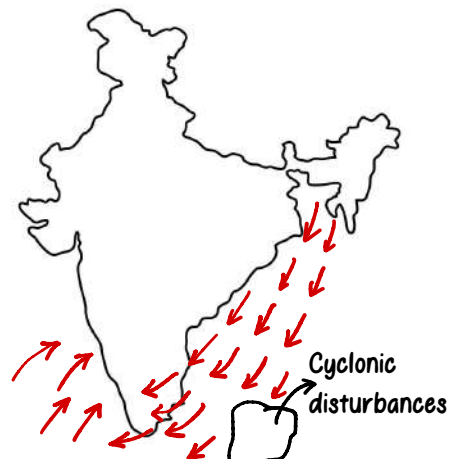
### Retreating monsoon

Also known as North-East Monsoon and it causes rainfall over Coromandel coast (Tamil Nadu/Andhra)

Flood in Chennai

### In North India

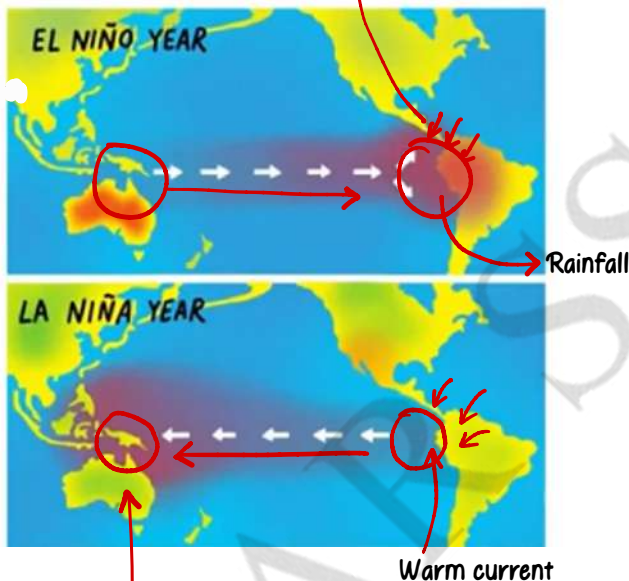
- Clear skies
- October heat: oppressing heat



## Effect of El-Nino in Monsoon

- Seen in 3-7 yrs
- Adversely affects the monsoon

North East winds weakens thus water returns

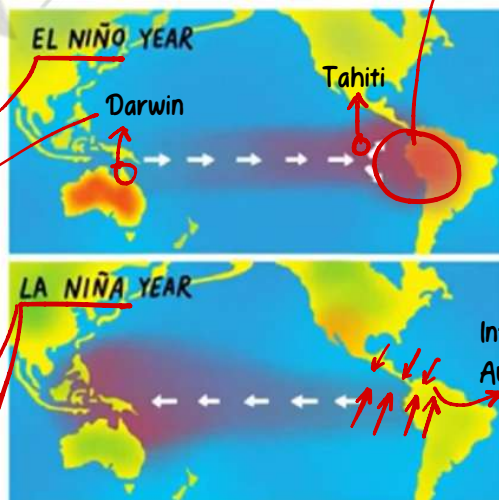


## Ocean Water

Surface water  
Deep water

Meaning: Child Christ/Small boy

- They measure the pressure difference b/w Tahiti and Darwin
- If the pressure difference is negative then El Niño will happen



Fishing Industry won't flourish  
No rain in Australia and India

Meaning: Small girl

Opposite to that of El Nino

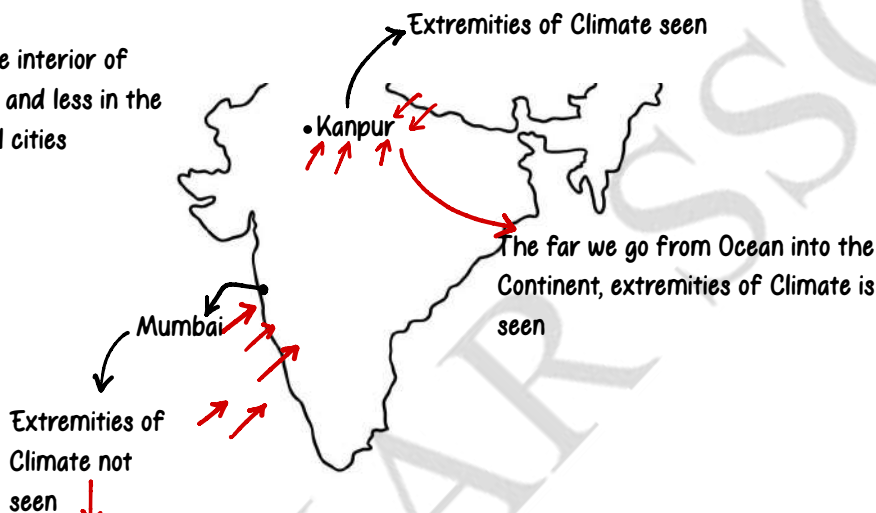
Intense rainfall in Australia and India



## Variation in Temperature/Rainfall

1. Diurnal Range of Temperature
2. Annual Range of Temperature
3. Annual Range of Rainfall

More in the interior of  
Continents and less in the  
the coastal cities



Because it is a coastal city.

Why?

1. Continentality effect
2. Moderating influence of Sea

## Seasons and Months

| Seasons  | Months<br>(According to the<br>Indian Calendar) | Months<br>(According to the<br>Indian Calendar) |
|----------|-------------------------------------------------|-------------------------------------------------|
| Vasanta  | Chaitra-Vaisakha                                | March-April                                     |
| Grishma  | Jyaistha-Asadha                                 | May-June                                        |
| Varsha   | Sravana-Bhadra                                  | July-August                                     |
| Sharada  | Asvina-Kartika                                  | September-October                               |
| Hemanta  | Margashirsa-Pausa                               | November-December                               |
| Shishira | Magha-Phalguna                                  | January-February                                |



## Forests

Eastern Himalayas

### 1. Evergreen Forests

- These are layered forests (shed their leaves not all together)
- They are seen in places of high temperature and high rainfall

30°C ↑

200 cm ↑

- In India, seen in Western side of Western Ghats, Andaman and Nicobar Islands, North-East
- Highest biodiversity seen here
- Eg: Amazon Rainforests (known as "Lungs of the World")

- Vegetation seen: Rosewood, Ebony, Mahogany, Cinchona, Aini, Epiphytes
- Also known as Desert covered by forest

### 2. Tropical Deciduous Forest

70-150 cm: Dry Deciduous

- Rainfall: 70 cm-200 cm

150-200 cm: Moist Deciduous

- Also known as Monsoon Forests
- These are the most predominant forests in India
- They are seen in: Peninsular Plateaus, North Indian Plains
- Trees:

Moist Deciduous: Shisham, Sagon, Sandalwood, Teak, Sal

Dry Deciduous: Tendu, Khair, Palas

Leaves are used to make Bidi

- Guttation: Hydathodes

### 3. Tropical Thorn Forests

- Rainfall: less than 50 cm
- Thorns are modified form of leaves (to avoid water loss)
- Trees: Babul, Khejri, Cactus
- Special type of grass seen here: Tussocky grass
- In India: seen in Rajasthan, Punjab, Gujarat





#### 4. Coniferous Forest

- Shape: Cone
- Seen in areas of high snowfall
- In India: seen in Upper Himalayas
- Trees:

Softwood trees: Chir, Pine, Cedar, Deodar, Spruce

↓  
Gymnosperms

target: 33% (India)

- National Forest Policy: 1952 → 1988
- Chipko Movement: 1978, led by Sundarlal Bahuguna
- Forest Day: 21st March
- Indian Forest Research Institute located in Dehradun, Uttarakhand

#### 5. Montane Forests

- In mountain region
- Categories:
  1. North India: seen in Himalayas → Bugyal Grassland seen
    - a. Upper part: Alpine/Coniferous m, eg: Rhododendron
    - b. Lower part: Deciduous forests
  2. South India

Evergreen forests seen here due to rainfall

Sholas seen here → Grassland/Forests of Western Ghats

India State Forest Report: tells about the total Forest cover in India

- Biennial Report
- Forest Cover: 21.71% → 2021 report
- Tree Cover: 2.9%
- Forest and Trees: 24.62%
- Highest Forest cover (area): Madhya Pradesh (1st), Arunachal Pradesh (2nd)
- Highest Forest cover (%): Mizoram (1st), Arunachal (2nd)
- Lowest Forest cover: Haryana
- Highest increase: Andhra Pradesh • Highest decrease: Arunachal > Manipur

#### 6. Mangrove Forests

- Known as Littoral/Swamp forests
- Seen in coastal areas
- More in Sundarban Deltas → Sundari Trees
- Region: West Bengal
- They are called living roots
- Trees are viviparous



Elephant grass but  
not nutritious

### Grasslands

Small grass  
but nutritious

#### Tropical grasslands

1. Savannah: in Africa
2. Campos grassland: Brazil
3. Llanos grassland: Venezuela

Known as Big Game country due to  
extreme poaching of animals

#### Temperate grasslands

1. Prairies: North America
2. Steppes: Asia/Europe
3. Velds: South Africa
4. Downs: Australia
5. Pustaz: Hungary
6. Canterbury: New Zealand
7. Pampas: Argentina

- Prairies: Known as wheat granaries of the world
- Pampas: grass seen "Alpha alpha grass" → nutrient rich grass

### Shifting Cultivation

- It is known as Slash and Burn Agriculture
- Not good for environment, causes deforestation and soil loses its productivity

### Across the world known by different names:

1. Indonesia: Ladang
2. Mexico: Milpa
3. Sri Lanka: Chena
4. Vietnam: Ray
5. Brazil: Roca
6. Venezuela: Konuko

### Across the India

1. Jhum: North East
2. Kumari: Western Ghats
3. Pama Dabi/Bringa: Odisha
4. Penda/Podu: Andhra Pradesh
5. Dipa: Chhattisgarh (Bastar district)
6. Waltre: Rajasthan
7. Kuruwa: Jharkhand
8. Bewar/Dahiya: Madhya Pradesh